DATA STRUCTURES PROJECT

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SECTION (CS-F)

Code Documentation

The code is composed of the following classes.

Main Class:

1. ringDHT

Classes:

1. c_LinkList
    a. *struct* CNode
2. FingerTable
    a. struct FNode
3. LinkList
    a. *struct* LNode
4. AVLTree
    a. *class* Node

Hashing Algorithm:

1. SHA-1

External Libraries:

1. boost

Utilities:

1. UtilityFunctions.h

ringDHT: The main class of the project. **Composed** of c_LinkList, holds functions to perform basic DHT Network operations such as uploading, fetching and deleting data. Holds function calls for Machine entry and leaving.

c_LinkList: A circular doubly linked list that holds the Machines as ***CNode*** as its nodes, therefore maintaining the basic structure of the entire Network. This class manages and caters all the fundamental operations of the ***ringDHT***.

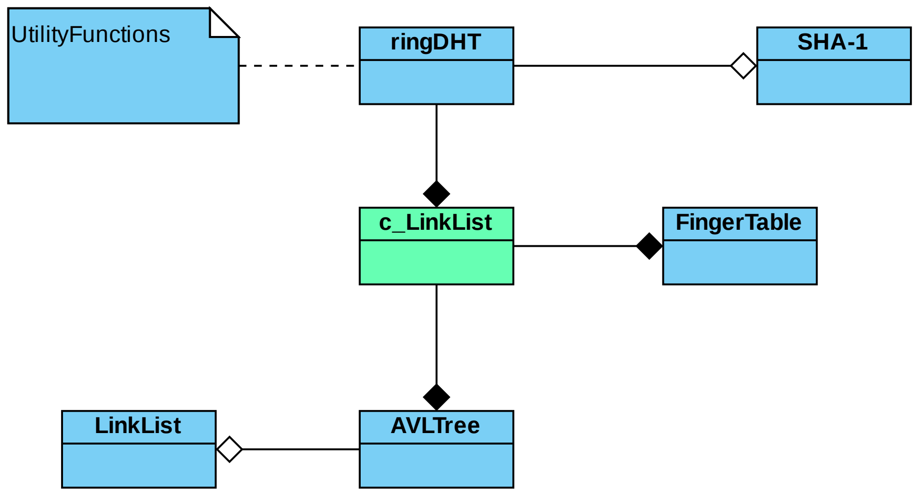
FingerTable: A class that simply holds the finger table for each Machine and its operations. Also composed of the ***CNode***.

CNode: A node struct holding objects of ***FingerTable*** and ***AVLTree***. Acts as nodes (machines) for the ***c_LinkList.***

LinkList: A class used to maintain a simple Link List Data Structure, composed of ***LNode***. This class helps in maintaining duplicate key, value pairs inside a Machine.

AVLTree: A class used to maintain a simple AVL Tree, composed of ***Node***.

Machine IPs and keys are hashed using **SHA-1** algorithm.
Additional utility functions are defined in **UtilityFunctions.h**


UtilityFunctions
ringDHT
SHA-1
c_LinkList
FingerTable
LinkList
AVLTree